

PP3 reference card

<http://pp3.sourceforge.net>

for version 1.3

General structure

```
# Some introductory comments,  
# i.e. what the file is about to do.
```

[Section I: Parameters.]

objects_and_labels

[Section II: Delete/add/modify objects and/or labels]

Both sections may be empty. If section II is empty, **objects_and_labels** should be omitted.

Section I: Parameters

The following top level keywords are allowed here: **filename**, **set**, **switch**, **color**, **line_width**, **line_style**, and **penalties**¹.

filename output orion.tex
writes the result to the file orion.tex. The file extension must be .tex and must not be omitted.

filename stars	stars.dat
filename nebulae	nebulae.dat
filename label_dimensions	labeldimens.dat
filename constellation_lines	lines.dat
filename boundaries	boundaries.dat
filename milky_way	milkyway.dat

This lets PP3 look for the respective data in the given file. The filenames must be full paths from the current directory.

filename latex_preamble mypreamble.tex
makes PP3 include the file mypreamble.tex with e. g. font parameters in every L^AT_EX run.

filename include myproject.pp3
includes myproject.pp3 that itself has a normal input script structure, except for the limitation that it mustn't include yet another file.

set center_rectascension 5.8
set center_declination 0
This sets the center of the printed map to the celestial coordinates (5^h48', 0°).

set box_width 15
set box_height 15
sets the dimensions of the printed map in centimetres. (15 for both is the default.)

set grad_per_cm 4
sets the scale of the map in degrees per centimetre. (4 is the default.)

set constellation ORI
tells PP3 to highlight the boundaries of constellation Orion. Must be in uppercase.

set shortest_constellation_line 0.1
makes PP3 suppress all constellation lines that are shorter than 0.1 centimetres. (This is the default.)

set label_skip 0.06
sets the distance between the outer rim of the celestial object and its label to 0.06 centimetres. (This is the default.)

set minimal_nebula_radius 0.1
means that all nebulae that would be smaller than 0.1 centimetres (in radius) are printed with a radius of *exactly* 0.1 cm. (This is the default.)

set faintest_cluster_magnitude 4.0
suppresses by default all stellar clusters that are fainter than 4.0^m. (This is the default.)

set faintest_diffuse_nebula_magnitude 8.0
suppresses by default all emission and reflection nebulae and all galaxies that are fainter than 8.0^m. (This is the default.)

set faintest_star_magnitude 7.0
suppresses by default all stars and all galaxies that are fainter than 7.0^m. (This is the default.)

set faintest_star_with_label_magnitude 3.7
means that for stars as bright as 3.7^m or brighter get automatically generated labels. (This is the default.)

set faintest_star_disk_magnitude 4.5
means that all stars that are fainter than 4.5^m are printed as mere dots instead of small circles. (This is the default.)

set minimal_star_radius 0.015
This sets the radius of the dots for all stars fainter than the limit determined by faintest_star_disk_magnitude to 0.015 centimetres. (This is the default.)

set star_scaling 2.0
makes all star circles two times bigger than normal. (Default is 1.0.)

set fontsize 12
sets the global font size to 12 pt. (Default is 10.)

switch milky_way	off
switch nebulae	on
switch grid	on
switch ecliptic	on
switch boundaries	on
switch constellation_lines	on
switch labels	on

This switches the display of the Milky Way off, and the nebulae, coordinate grid, ecliptic, constellation boundaries, boundaries of the highlighted constellation, and all labels on. This is the default.

switch eps_output **on**
switch pdf_output **on**

The first line makes PP3 generating an EPS file from the map via a dvips call, the second creates a PDF file additionally by calling ps2pdf.

¹not explained on this reference card

switch colored_stars **off**

This disables PP3’s default behaviour to try to illustrate the star’s actual colour in the map. Instead, all stars get the same colour explicitly defined by “**color** stars”.

color background	0	0	0.4
color grid	0	0.298	0.447
color ecliptic	1	0	0
color boundaries	0.5	0.5	0
color highlighted_boundaries	1	1	0
color constellation_lines	0	1	0
color milky_way	0	0	1
color nebulae	1	1	1

This sets the colours of these map elements to the respective RGB colour. The colour values range from 0 to 1. What you see here are the default values.

color stars	1	1	1
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This sets the colour of stars to white (the default), however only if you switch colored_stars off.

color labels	0	1	1
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sets the colour of all automated labels to cyan (the default).

color text_labels	1	1	0
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sets the colour of all user defined labels to yellow (the default).

line_width grid	0.025
line_width ecliptic	0.018
line_width boundaries	0.035
line_width highlighted_boundaries	0.035
line_width nebulae	0.018
line_width constellation_lines	0.035

This sets the line widths of these map elements to the respective width in centimetres. What you see here are the default values.

line_style grid	solid
line_style ecliptic	dashed
line_style boundaries	dashed
line_style highlighted_boundaries	dashed
line_style nebulae	solid
line_style constellation_lines	solid

This sets the line styles of these map elements to the respective style. What you see here are the default values.

Section II: Objects and labels

The following top level keywords are allowed here: **add**, **delete**, **reposition**, **add_labels**, **delete_labels**, **set_label_text**, and **text**.

add IC 5070 NGC 7000 ;

adds the nebulae IC 5070 and NGC 7000 to the map with their labels.

delete LEO 63 HD 97605 ;

removes the two stars 63 Leo and HD 97605 from the map.

reposition LEO 32 SE ;

positions the label of 32 Leo to the lower right (SE = South East).

add_labels M 66 LEO 24 ;

adds the labels of M 66 and 24 Leo to the map. The positioning is still automatic.

delete_labels M 43 ;

deletes the label of the nebula M 43 from the map, *not* the nebula itself.

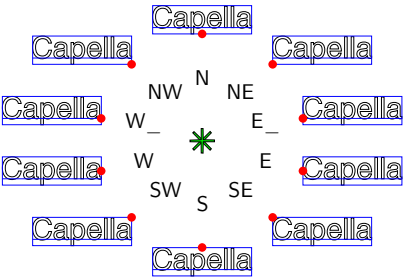
set_label_text UMA 50 “\footnotesize_Dubhe”

replaces the default label text for 50 UMa with “\footnotesize Dubhe”.

text “\small Wolf 359\hskip0.3em
 \psdots[dotstyle=+,dotangle=45](0,0)”
 at 10.902 7.32
 color 0.3 0.3 0.9333 **towards** NW ;

This prints the L^AT_EX commands between the “...” at the celestial point (10^h54^m7.2^s, 7.32°), in a grey-blue colour, and to the upper left of the point.

This graphics shows the possible values for **reposition** and the **towards** option (the large dots lie exactly on the celestial position given after the **at** option):



text “\$#3\$” **at** 0 20 **along** declination
 tics rectascension 1 **towards** N ;
 text “\$#5\$” **at** 11 0 **along** declination
 tics declination 10 **towards** S ;

This creates automatically generated labels for the 20^o declination circle (every whole hour), and for the 11^h rectascension line (every 10 declination degrees). The following placeholders are available:

- #1 Rectascension in hours.
- #2 Declination in degrees.
- #3 Rectascension in integer hours (truncated, not rounded).
- #4 Rectascension fraction of hour in minutes (truncated, not rounded).
- #5 Declination in rounded integer degrees.

L^AT_EX preamble

\usepackage{amsmath}
\usepackage[T1]{fontenc}
\usepackage{eulervm}
\renewcommand*{sfdefault}{phv}
\AtBeginDocument{\sffamily\boldmath}

This sets some global font parameters: We use a sans serif font (Helvetica) as the standard, and Euler for all Greek letters.

\usepackage{relsize}
\renewcommand*{\Messier}[1]{%
 \footnotesize\smaller M}\, #1}
\renewcommand*{\NGC}[1]{%
 \footnotesize\smaller NGC}\, #1}
\renewcommand*{\IC}[1]{%
 \footnotesize\smaller IC}\, #1}

This resets the catalogue hooks. \Messier, \NGC, and \IC take one parameter which is the catalogue number. In this example they print the catalogue abbreviation a little bit smaller.

\renewcommand*{\Starname}[1]{#1}
\renewcommand*{\Label}[1]{#1} % automatic labels
\renewcommand*{\TextLabel}[1]{#1} % user labels
\renewcommand*{\FlexLabel}[1]{\bfseries #1}
\renewcommand*{\TicMark}[1]{\mdseries
 \scriptsize\mathversion{normal} #1}
\renewcommand{\DP}{.}

Further hooks: This makes all flexes bold and all tics small and non-bold. \DP is the decimal point (the dot is the default). In the first three lines you see default definitions: The parameter is just mapped to the output.